

Question 1

```
[ec2-user@ip-172-31-3-136 ~]$ sudo dnf install docker -y
Amazon Linux 2023 Minimal Livepatch repository
Amazon Linux 2023 Kernel Livepatch repository
Last metadata expiration check: 0:00:03 ago on Sat Jun 13 06:16:31 2026.

Dependencies resolved.
=====
Package                               Architecture      Version           Repository        Size
Installing:
docker                               x86_64            25.0.16-1.amzn2023.0.2   amazonlinux        46 M
Installing dependencies:
container-selinux                    x86_64            4:2.245.0-1.amzn2023.0.1   amazonlinux        58 k
containerd                           x86_64            2.2.4-1.amzn2023.0.1     amazonlinux        24 M
iptables-libse                        x86_64            1.8.6-3.amzn2023.0.2     amazonlinux        401 k
iptables-nft                         x86_64            1.8.6-3.amzn2023.0.2     amazonlinux        183 k
libnftables                          x86_64            3.0-1.amzn2023.0.1       amazonlinux        75 k
libnftables-conntrack                x86_64            1.0.6-2.amzn2023.0.2     amazonlinux        58 k
libnftables-link                     x86_64            1.0.1-15.amzn2023.0.2    amazonlinux        30 k
libnftnl                             x86_64            1.2.2-2.amzn2023.0.2     amazonlinux        84 k
nftables                             x86_64            2.3-1.amzn2023.0.4       amazonlinux        15 k
runc                                  x86_64            1.3.5-1.amzn2023.0.2     amazonlinux        3.9 M
Transaction Summary
-----
Total download size: 76 M
Total installed size: 204 M
Downloading Packagets:
(1/11): container-selinux-2.245.0-1.amzn2023.0.1.rpm                  1.6 MB/s | 58 kB | 00:00
(2/11): iptables-libse-1.8.6-3.amzn2023.0.2.x86_64.rpm              15 MB/s | 401 kB | 00:00

I-030b3a727d25acfb8 (practicalinstance)
PublicPip: 3.109.1.134 PrivatePip: 172.31.3.136
```

1.

```
[ec2-user@ip-172-31-3-136 ~]$ sudo systemctl start docker
[ec2-user@ip-172-31-3-136 ~]$ sudo systemctl status docker
docker version 25.0.16, build 0b0097
[ec2-user@ip-172-31-3-136 ~]$ sudo docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55986776bd: Pull complete
Digest: sha256:608a100c71651bf5b773c89083b4a1ad7ef4b2bd05d7a7e552271e03123692ad
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (wasid)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently seeing.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

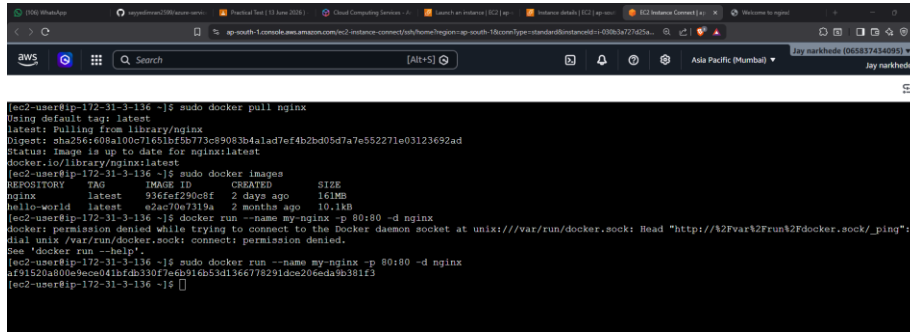
[ec2-user@ip-172-31-3-136 ~]$
```

2.

```
[ec2-user@ip-172-31-3-136 ~]$ sudo docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
Digest: sha256:608a100c71651bf5b773c89083b4a1ad7ef4b2bd05d7a7e552271e03123692ad
Status: Image is up to date for nginx:latest
docker.io/library/nginx:latest
[ec2-user@ip-172-31-3-136 ~]$ sudo docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
nginx         latest   936fef290c8f   2 days ago    161MB
hello-world   latest   e2ac70e7319a   2 months ago  10.1kB
[ec2-user@ip-172-31-3-136 ~]$
```

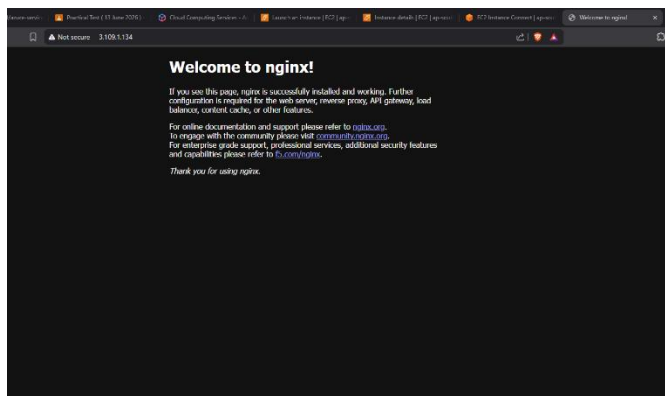
3.

4. Running nginx



```
[ec2-user@ip-172-31-3-136 ~]$ sudo docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
Digest: sha256:608a100c71651bf5b773c89083b4a1ad7ef4b2bd05d7a7e552271e03123692ad
Status: Image is up to date for nginx:latest
docker.io/library/nginx:latest
[ec2-user@ip-172-31-3-136 ~]$ sudo docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
nginx         latest   936ca29b0c8f   2 days ago    161MB
hello-world   latest   e2a070e7319a   2 months ago  10.1kB
[ec2-user@ip-172-31-3-136 ~]$ docker run --name my-nginx -p 80:80 -d nginx
docker: permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Head "http://x2Fvart2Frunx2Fdocker.sock/_ping":
dial unix /var/run/docker.sock: connect: permission denied.
See 'docker run --help'.
[ec2-user@ip-172-31-3-136 ~]$ sudo docker run --name my-nginx -p 80:80 -d nginx
af91520a800e9ece041bfdb330f7edb916b53d136e778291dce206eda9b381f3
[ec2-user@ip-172-31-3-136 ~]$
```

5. Accessing the page



6.

Question 2

1.

```
aws [Alt+S]

[ec2-user@ip-172-31-3-136 ~]$ sudo docker volume create student data
"docker volume create" requires at most 1 argument.
See 'docker volume create --help'.

Usage:  docker volume create [OPTIONS] [VOLUME]

Create a volume
[ec2-user@ip-172-31-3-136 ~]$ sudo docker volume create studentdata
studentdata
```

2.

```
[ec2-user@ip-172-31-3-136 ~]$ sudo docker volume create student data
"docker volume create" requires at most 1 argument.
See 'docker volume create --help'.

Usage:  docker volume create [OPTIONS] [VOLUME]

Create a volume
[ec2-user@ip-172-31-3-136 ~]$ sudo docker volume create studentdata
studentdata
[ec2-user@ip-172-31-3-136 ~]$ ls
[ec2-user@ip-172-31-3-136 ~]$ sudo docker run -it -v studentdata:/ubuntu bash
Unable to find image 'bash:latest' locally
latest: Pulling from library/bash
9b70e313681f: Pull complete
15249c1264b1: Pull complete
87abf3302012: Pull complete
Digest: sha256:27639cc1442e5d2ed9778c122a60efff06f73812c0320e55fa8e5e0ec1f4f26
Status: Downloaded newer image for bash:latest
bash-5.3#
```

3.

```
aws [Alt+S] Ask Amazon Q

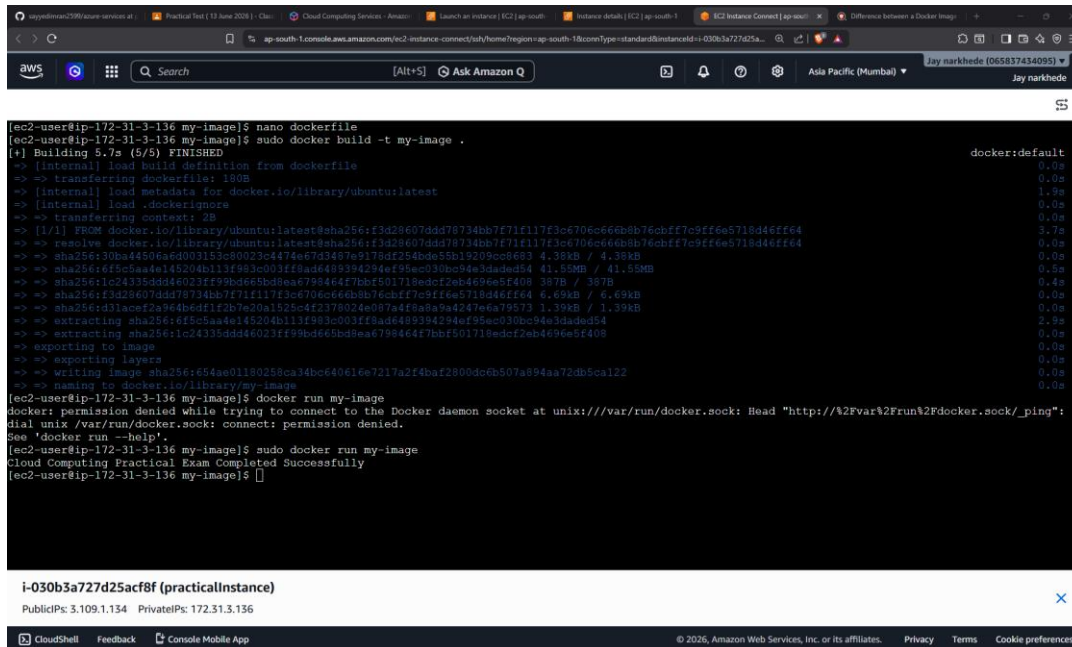
bash-5.3# echo "name: jay narkhede, roll no. : 24104A0038" > /data/info.txt
bash-5.3#
```

4.

```
aws [Alt+S] Ask Amazon Q

bash-5.3# echo "name: jay narkhede, roll no. : 24104A0038" > /data/info.txt
bash-5.3# cat /data/info.txt
name: jay narkhede, roll no. : 24104A0038
bash-5.3#
```

Question 3



```
[ec2-user@ip-172-31-3-136 my-image]$ nano dockerfile
[ec2-user@ip-172-31-3-136 my-image]$ sudo docker build -t my-image .
[+] Building 5.7s (5/5) FINISHED
=> [internal] load build definition from dockerfile
=> => transferring dockerfile: 190B
=> [internal] load metadata for docker.io/library/ubuntu:latest
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/1] FROM docker.io/library/ubuntu:latest@sha256:f3d28607ddd78734bb7f71f117f3c6706c666b8b76cbff7c9ff6e5718d46ff64
=> => resolve docker.io/library/ubuntu:latest@sha256:f3d28607ddd78734bb7f71f117f3c6706c666b8b76cbff7c9ff6e5718d46ff64
=> => sha256:30ba44506a6d003183c80023c4474ee7d3487e9178df254bde5b19209c0e6e3 4.38kB / 4.38kB
=> => sha256:6f5c5ba4e145204b113f993c003f1f9ad6489394294ef95ec030bc94e3daded54 41.50MB / 41.50MB
=> => sha256:1c24335ddd46023f1f99bd666b8b8eae798464f7bbf5b1718edcf2eb4e96e5f408 357B / 357B
=> => sha256:f3d28607ddd78734bb7f71f117f3c6706c666b8b76cbff7c9ff6e5718d46ff64 6.69kB / 6.69kB
=> => sha256:d31acef2a94b6d1f2b7e20a1525c4f2378024e087a4f8a8a9a42476ea79573 1.39kB / 1.39kB
=> => extracting sha256:6f5c5ba4e145204b113f993c003f1f9ad6489394294ef95ec030bc94e3daded54 2.99
=> => extracting sha256:1c24335ddd46023f1f99bd666b8b8eae798464f7bbf5b1718edcf2eb4e96e5f408 0.0s
=> => exporting to image 0.0s
=> => exporting layers 0.0s
=> => writing image sha256:654ae01180258ca34bc640616e7217a2f4baf2800dc6b507a894aa72db5ca122 0.0s
=> => naming to docker.io/library/my-image 0.0s
[ec2-user@ip-172-31-3-136 my-image]$ docker run my-image
docker: permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Head "http://%2Fvar%2Frun%2Fdocker.sock/_ping":
dial unix /var/run/docker.sock: connect: permission denied.
See 'docker run --help'.
[ec2-user@ip-172-31-3-136 my-image]$ sudo docker run my-image
Cloud Computing Practical Exam Completed Successfully
[ec2-user@ip-172-31-3-136 my-image]$
```

i-030b3a727d25acf8f (practicalInstance)
PublicIP: 3.109.1.134 PrivateIP: 172.31.3.136

Bonus question

1. Docker images are read only template
docker container is a mutable instance made form docker image. It is isolated.
2. Docker volumes are used to **persist data** outside the container's lifecycle. Since containers data is deleted after removal of container. Volumes provide the storage.